MILITARY SPECIFICATION

INSULATING COMPOUND, ELECTRICAL, LOW VISCOSITY, SILICONE RUBBER

This specification is approved for use by the US Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a two component, room temperature curing, low viscosity, white, silicone rubber **insulating compound.**

2. APPLICABLE DOCUMENTS

* 2.1 Issues of Documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

* MIL-C-47003	Curing Agent, Silicone Rubber

STANDARDS

Federal

FED-STD-406	Plastics, Methods of Testing
FED-STD-601	Rubber, Sampling and Testing

Beneficial Comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: COMMANDER, US Army Missile Command, ATTN: DRSMI-RSDS, Redstone Arsenal, AL 35898 by using the self-addressed Standardization Document Improvement Proposal (DD FORM 1426) appearing at the end of this document or by letter.

FSC 5970

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	MIL-STD-129	Marking for Shipment and Storage
*	MIL-STD-1188	Commercial Packaging of Supplies and Equipment

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The folowing documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposals shall apply.

* American Society for Testing and Materials

ASTM D 1084-63	Standard Methods of Tests for Viscosity of Adhesives

(Application for copies should be addressed to the American Society for Testing Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

* Institute of Printed Circuits

IPC-TM-650	Glass Transition Temperature (TMA Method)

(Copies of IPC publications may be obtained from the Institute of Printed Circuits, 1717 Howard Street, Evanston, IL 60202.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

* 3.1 Preproduction sample. Unless otherwise specified (see 6.2), a preproductions sample (4.2) of silicone rubber **insulating compound shall** meet the requirements of this specifications.

* 3.2 Material. The material shall consist of a low viscosity, white silicone rubber base resin capable of being catalyzed at room temperature with a curing agent comforming to MIL-C-47003.

3.3 Electrical properties. The cured material shall meet the requirements specified in Table I at 25 degrees Celsius (C) (77 degrees Fahrenheit (F)).

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Table I. Electrical Properties

Property	Values
Dielectric Strength, Volts Per mil	300, minimum
Volume Resistivity, ohm-centimeter	1×10^{12}, minimum
Dielectric Constant, 60 to 10^8 cycles	5.0, maximum

3.4 Mechanical Properties. The cured material shall meet the requirements specified in Table II.

Table II. Mechanical Properties

Property	Value
Elongation, percent	
at 25 degrees Celsius (C) (77 degrees Fahrenheit (F))	100, minimum
at 232 degrees Celsius (C) (450 degrees Fahrenheit (F)	75, minimum
Tensile Strength, pounds per square inch	250, minimum
at 25 degrees Celsius (C) (77 degrees Fahrenheit (F))	160, minimum
at 232 degrees Celsius (C) (450 degrees Fahrenheit (F))	
Tear Resistance, pounds per inch	
at 25 degrees Celsius (C) (77 degrees Fahrenheit (F))	10, minimum
Low temperature resistance	
at -54 degrees Celsius (C) (-65 degrees Fahrenheit (F))	No cracks, checks or breaks

3.5 Physical properties.

3.5.1 Viscosity. The viscosity of the base resin compound shall be 80 to 180 poises.

3.5.2 Specific gravity. The specific gravity of the cured compound shall be 1.18 plus or minus 0.04.

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3.5.3 Hardness. The Shore A hardness of the cured compound at 25 degrees Celsius (C) (77 degrees Fahrenheit (F)) shall be within the range of 40 to 55.

3.5.4 Shelf Life. The shelf life of the material in unopened containers shall be a minimum of six months from date of manufacturer when stored at temperature less than 27 degrees Celsius (C) (80.6 degrees Fahrenheit (F)).

3.6 Glass transition temperature (Tg). To be capable of meeting the requirements of this specification, the apparent Tg or softening point shall be as specified by the manufacturer. The test specimen shall be only correctly mixed, cured, and aged per the manufacturer's instruction. The test method shall be as specified in 4.7 or any other method that gives comparable results. (See 6.3)

3.7 Workmanship. The workmanship shall be such as to insure a product which is uniform and in conformance with this specification. The silicone rubber base resin shall be free of dirt, foreign material or other contaminants.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction sample. The preproduction sample shall be prepared using the same methods proposed for the preparation of subsequent production lots of material. The preproduction sample shall be subjected to all examinations and tests specified herein. Preproduction samples which do not meet all the requirements of this specification shall be rejected and returned to the contractor. Subsequent quantities will not be considered for acceptance until approval of the preproduction sample has been obtained.

4.3 Classification of examination and tests. The examination and testing of material shall be classified as follows:

a. Preproduction tests (see 4.4).

b. Quality conformance tests (see 4.5).

4.4 Preproduction tests. Preproduction tests shall be conducted only on the preproduction sample and shall consist of all the examinations and tests specified herein.

* 4.5 Quality conformance tests. Quality conformance tests for acceptance of material shall consist of the following examinations and tests:

a. Workmanship (see 3.7).

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b. Viscosity (see 4.6.4.1).

c. Specific gravity (see 4.6.4.2).

d. Hardness (see 4.6.4.3).

4.5.1 Lot size and sampling.

4.5.1.1 Lot size. Lot size shall consist of all the base resin manufactured at one time from one batch, forming part of one contract or order, and submitted for acceptance at the same time and place.

4.5.1.2 Sampling. Unless otherwise specified (see 6.2), one quart from one unit representative of each lot (see 4.5.1.1) shall be selected at random for quality conformance testing. Each container in lot shall be considered as a unit of product. Failure of sample to meet the quality conformance test requirements shall be cause for lot rejection.

4.6 Test methods.

4.6.1 Test specimen preparation. The test specimens shall be cured for 72 to 76 hours plus or minus 3 degrees Celsius (C) (73.4 plus or minus 5.4 degrees Fahrenheit (F) after removal from mold. The elongation and tensile strength specimens for elevated temperature testing shall be conditioned for an additional 68 to 70 hours at 232 plus or minus 3 degrees Celsius (C) (450 plus or minus 5.4 degrees Fahrenheit (F).

4.6.2 Electrical property tests. The electrical property tests shall be conducted in accordance with FED-STD-406 and the applicable test method specified herein.

4.6.2.1 Dielectric strength. The dielectric strength specified in Table I shall be determined in accordance with Method 4031.

4.6.2.2 Volume resistivity. The volume resistivity specified in Table I shall be determined in accordance with Method 4041.

4.6.2.3 Dielectric constant. The dielectric constant specified in Table I shall be detemined in accordance with Method 4021.

4.6.3 Mechanical property tests. The mechanical property tests shall be conducted in accordance with FED-STD-601 and the applicable test method specified herein.

4.6.3.1 Elongation. The elongation specified in Table II shall be determined in accordance with Method 4121 using same specimen required for tensile strength test (see 4.6.3.2).

4.6.3.2 Tensile strength. The tensile strength specified in Table II shall be determined in accordance with Method 4111, using die number III for test specimen configuration.

4.6.3.3 Tear resistance. The tear strength specified in Table II shall be determined in accordance with Method 4211.

4.6.3.4 Low temperature resistance. The low temperature resistance specified in Table II shall be determined in accordance with Method 5311.1 after conditioning at the specified temperature for 30 minutes minimum.

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4.6.4 Physical property tests.

4.6.4.1 Viscosity. The viscosity specified in 3.5.1 shall be determined in accordance with ASTM D 1084, Method B, using a Brookfield model RVF, number 5 spindle at 4 revolutions per minute, or equivalent. The 100 scale shall be used and 10 revolutions shall be allowed before a reading is taken.

4.6.4.2 Specific gravity. The specific gravity specified in 3.5.2. shall be determined in accordance with FED-STD-601, Method 14011.

4.6.4.3 Hardness. The hardness specified in 3.5.3 shall be determined in accordance with FED-STD-601, Method 3021.

4.6.4.4 Shelf life. The base resin compound shall be tested for conformance with all the requirements specified herein after completion of storage period. The manufacturer's certificate of compliance with the shelf life requirement (see 3.5.4) may be accepted during the storage period.

* 4.7 Glass transition temperature test. The apparent Tg or softening point temperature shall be determined in accordance with method 2.4.2.4 of IPC-TM-650 or a thermal analysis (TMA) method (see 6.3) that utilizes a penetration probe. Standard loading rate, temperature rate, coolant and gas purge as operating parameters shall be stated and remain the same for all tests. The resultant Tg shall conform to that specified in accordance with 3.6.

NOTE: Thermomechanical analysis test equipment manufactured by the Perkin-Elmer TMS-1 or the I. E. DuPont 941 TMA.

4.8 Packaging. Packaging shall be examined for compliance with Section 5.

5. PACKAGING.

* 5.1 Commercial.

* 5.1.1 Preservation, packing, unitization and marking. Preservation, packing, unitization and marking shall be in accordance with MIL-STD-1188. In addition, shelf life markings are required.

5.2 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with requirements of MIL-STD-129.

6. NOTES

* 6.1 Intended use. The material is intended for use where a low viscosity silicone rubber insulating compound capable of curing at room temperature is needed for electronic potting, encapsulating, and molding application. A primer, conforming to MIL-P-47170, should be used to promote maximum adhesion of insulating compound to materials other than silicone rubber.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a preproduction sample is required (see 3.1), and if so, pertinent details.

c. Sampling plan, if other than specified (see 4.5.1.2).

d. Any special marking requirements (see 5.2).

e. Container size (see 5.1).

* 6.3 Thermomechanical analysis (TMA). Thermomechanical analysis (TMA) is a thermal technique for characterization and quality control of encapsulating and potting compounds. By measuring dimensional changes (expansion mode) or viscoelastic changes (penetration mode) properties such as coefficient of expansion modulus, heat distortion and softening point can be determined precisely and rapidly on small samples in any form-powder, pellet, film, fiber or molded part. The measurements correlate well with the results of many ASTM physical tests. Compared with the ASTM test method, the TMA method is very rapid, requires a small amount of sample, is nondestructive to the manufactured item and the test equipment outlay is inexpensive. The Tg or softening point will vary or differ from one manufacturer to another according to the additives, modifiers, plasticizers, and catalysts. All of these variables cause changes in the cross linking of the polymer.

6.4 Changes to previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER AND TITLE

MIL-I-47199A(MI) INSULATING COMPOUND, ELECTRICAL, LOW VISCOSITY, SILICONE RUBBER

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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☐ YES ☐ NO *(If "Yes", in what way?)*

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